

Commit0269252

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 ActuallyNull committed last month

quick commit since im leaving town for a bit, added some more class weight for other arrhythmia, and extra augmentation for other arrhythmia class since thats the biggest problem so far

 main

1 parent3cecf59commit0269252



Filter files...

data_processing.py

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2 files changed+23-1 lines changed

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data_processing.py

+22-1



@@ -80,6 +80,7 @@ def preprocess_dataset(physionet_path, reference_file_path, X, y, fs=300):

8080X_balanced, y_balanced = smote_resampling(X_filtered, y_processed)

8181return X_balanced, y_balanced

8282

83+

8384def create_train_val_test_splits(fs=300):

8485X = []

8586y = []

@@ -104,11 +105,12 @@ def create_train_val_test_splits(fs=300):

104105X_val, X_test, y_val, y_test = train_test_split(

105106X_temp, y_temp, test_size=0.5, random_state=42, stratify=y_temp

106107)

108+

109+# normal augmentation

107110batch_size = 500 # Adjust as needed for your RAM

108111X_train_aug = []

109112for i in range(0, len(X_train), batch_size):

110113batch = X_train[i:i+batch_size]

111-# For strong augmentation, optionally mixup with other signals in the batch

112114batch_aug = []

113115for x in batch:

114116if augmentor.strong and np.random.rand() < 0.2 and len(batch) > 1:

@@ -121,7 +123,26 @@ def create_train_val_test_splits(fs=300):

121123batch_aug = np.expand_dims(np.array(batch_aug, dtype=np.float32), axis=-1)

122124X_train_aug.append(batch_aug)

123125X_train_aug = np.concatenate(X_train_aug, axis=0)

126+

127+# extra augmentation for class 2 cause it has more trouble

128+other_class_idx = le.transform(["Other Arrhythmia"])[0]

129+other_mask = (y_train == other_class_idx)

130+X_other = X_train[other_mask]

131+y_other = y_train[other_mask]

132+

133+extra_augmented = []

134+for x in X_other:

135+for _ in range(3): # triple augmentations for more variability

136+extra_augmented.append(augmentor.augment(x.squeeze()))

137+extra_augmented = np.expand_dims(np.array(extra_augmented, dtype=np.float32), axis=-1)

138+y_other_extra = np.full(len(extra_augmented), other_class_idx)

139+

140+# Append extra augmented "Other Arrhythmia" samples

141+X_train_aug = np.concatenate([X_train_aug, extra_augmented], axis=0)

142+y_train = np.concatenate([y_train, y_other_extra], axis=0)

143+

124144# Clean up memory

125145del X, y, X_processed, X_cnn, X_train, X_temp, batch, batch_aug

126146gc.collect()

147+

127148return X_train_aug, y_train, X_val, y_val, X_test, y_test, le



model.py

+1



@@ -24,6 +24,7 @@

2424

2525weights = compute_class_weight('balanced', classes=np.array([0,1,2]), y=y_train)

2626class_weights = dict(enumerate(weights))

27+class_weights[2] *= 1.3 # give more weight to other arrhythmia since it gets misclassified a lot more

2728

2829callback = [

2930EarlyStopping(monitor='val_loss', patience=7, verbose=1, restore_best_weights=True),

⋮

Comments0



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